

WEEK 4 – POWER BI EDA

Shopify Stock – Volume and SMA Analysis

Exploratory Data Analysis using Power BI

1. Introduction

In Week 4, I worked on Exploratory Data Analysis (EDA) using Power BI with Shopify stock data. The main focus of this project was to analyse trading volume and understand the behaviour of the 20-day and 50-day Simple Moving Averages (SMA). I also created a Date table, connected it with the stock data and prepared an interactive report page.

2. Objectives

The main objectives of my Week 4 project were to calculate Simple Moving Averages using DAX, compare short-term and long-term volume trends, and create interactive Power BI visuals.

- Create a Date table for time-based analysis.
- Create a 20-day Simple Moving Average for trading volume.
- Create a 50-day Simple Moving Average for trading volume.
- Create moving-average comparison logic using DAX.
- Create visuals for trading volume and SMA analysis.
- Add a date range slicer for interactive analysis.
- Create a final dashboard for Shopify stock volume analysis.

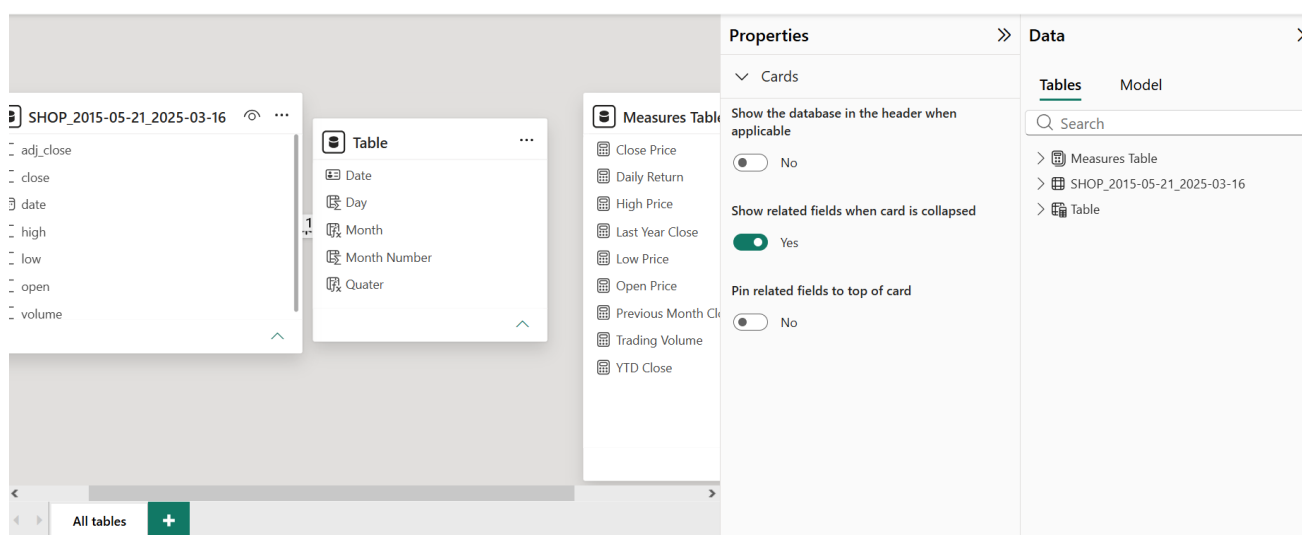
3. Dataset Used

For this project, I used the Shopify stock dataset. The dataset contains historical stock information such as date, open price, high price, low price, close price, adjusted close and trading volume. The Volume column was mainly used for calculating the moving averages.

4. Creating the Date Table

First, I created a separate Date table in Power BI using the minimum and maximum dates available in the Shopify stock data. The Date table contains Date, Day, Month, Month Number and Quarter fields. I then created a relationship between the Date table and the Shopify stock table. This helped me use the date field correctly for time-based calculations and visuals.

Figure 1: Date table, stock table and Measures Table



5. Creating the 20-Day SMA

After creating the Date table, I created a DAX measure to calculate the 20-day Simple Moving Average of trading volume. The calculation uses the current date and takes the average volume over the selected period.

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE(Shopify_Stock[Volume]),  
        DateInPeriod  
    )
```

6. Creating the 50-Day SMA

Next, I created the 50-day Simple Moving Average using the same approach. The difference is that the calculation uses a 50-day period. This gives a smoother view of the longer-term trading-volume trend.

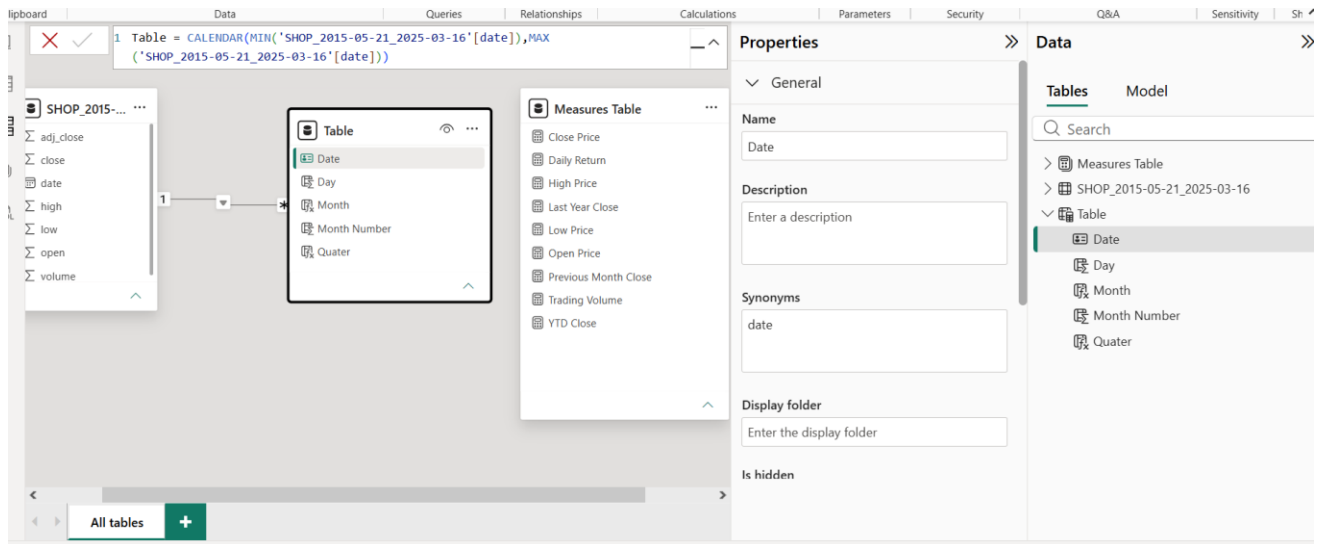
```
Volume 50 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE(Shopify_Stock[Volume]),  
        DateInPeriod  
    )
```

7. Moving Average Crossover

I also created a moving-average crossover measure to compare the 20-day SMA with the 50-day SMA. When the 20-day SMA is greater than the 50-day SMA, the result is shown as Bullish. Otherwise, the result is shown as Bearish. This measure helps compare the short-term and longer-term volume trends.

```
MA Crossover Signal =  
IF(  
    ISBLANK([20 SMA]) || ISBLANK([50 SMA]),  
    BLANK(),  
    IF(  
        [20 SMA] > [50 SMA],  
        "Bullish (20 > 50)",  
        "Bearish (20 < 50)"  
    )  
)
```

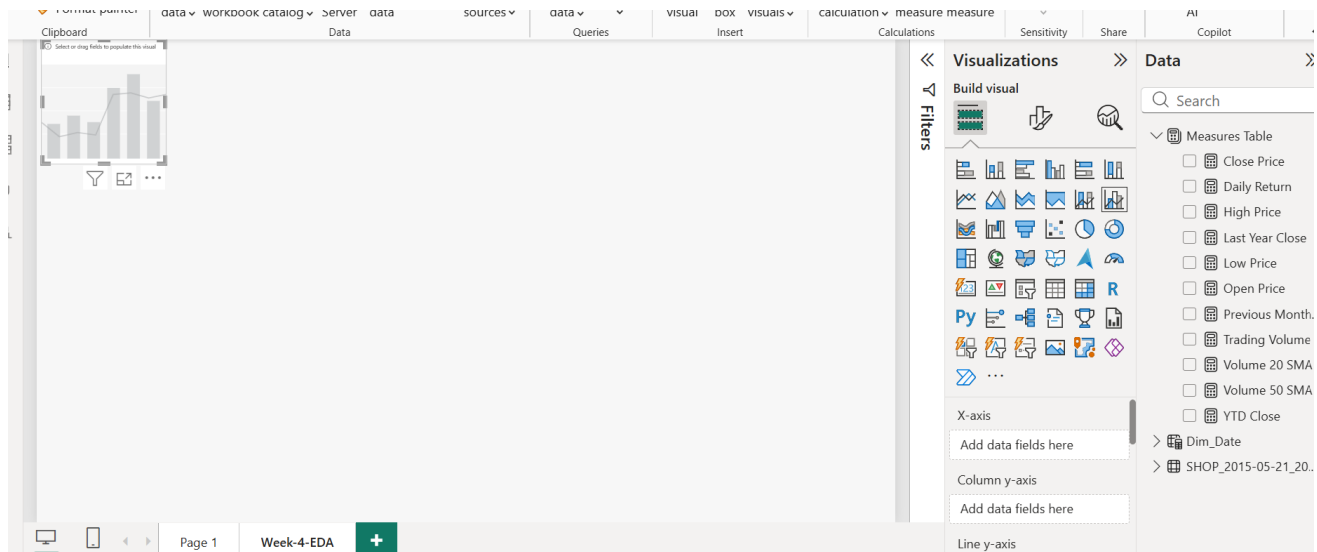
Figure 2: Date table relationship and measures



8. Creating the Visualizations

After creating the DAX measures, I moved to the Report view and created the Week-4-EDA report page. I used Power BI visuals to display the trading volume and moving-average information. The visualizations were arranged so that the volume trend and SMA comparison could be viewed easily.

Figure 3: Week-4-EDA report page while creating the visuals



9. Adding the Date Range Slicer

I added a Date slicer to the report so that the analysis can be viewed for a selected period. The slicer uses the Date field from the Date table and allows the user to select the required date range. This makes the dashboard more interactive.

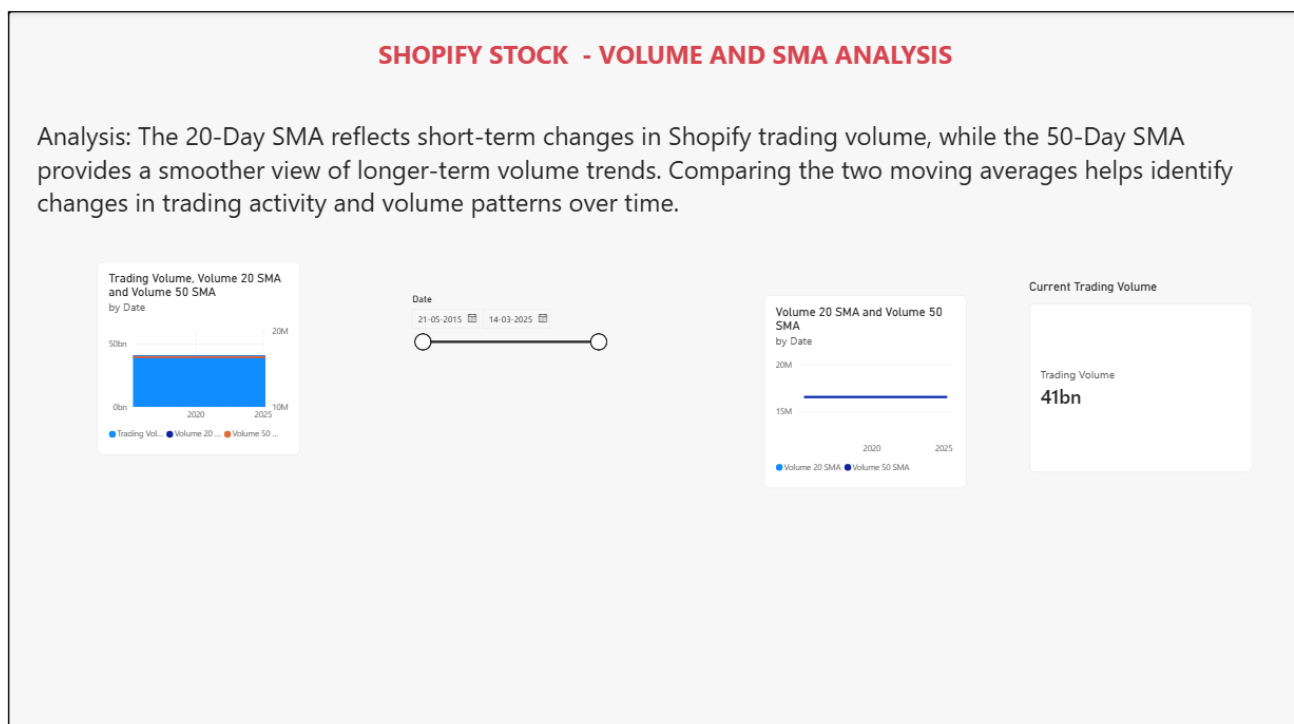
Figure 4: Trading volume, SMA visuals and date range slicer



10. Final Dashboard

Finally, I arranged the visuals into a simple Shopify Stock Volume and SMA Analysis dashboard. The dashboard contains an analysis description, trading-volume visual, 20-day and 50-day SMA comparison, date range selection and a Current Trading Volume card. The dashboard provides a clear view of trading activity and moving-average trends.

Figure 5: Final Shopify Stock – Volume and SMA Analysis dashboard



11. Analysis

From the dashboard, I observed that the 20-day SMA provides a shorter-term view of trading volume, while the 50-day SMA provides a smoother longer-term view. Comparing these two measures helps in

understanding changes in trading activity over time. The date slicer also makes it possible to focus on a particular period.

12. Work Completed in Week 4

In this Week 4 work, I completed the Date table, the relationship with the stock data, the 20-day SMA, the 50-day SMA, the moving-average comparison, the trading-volume visuals, the date range slicer and the final dashboard shown in the screenshots.

The Week 4 instructions also mention additional features such as an OHLC price trend chart, conditional volume colours, a field-parameter metric selector and a custom Stock_Tooltip page. These can be added as further improvements to the report.

13. Conclusion

In conclusion, I used Power BI to perform Exploratory Data Analysis on Shopify stock data. I created a Date table and used DAX to calculate 20-day and 50-day Simple Moving Averages for trading volume. I then created interactive visuals and a date slicer to analyse the volume trends. The final dashboard gives a simple and clear view of Shopify stock trading volume and its moving-average trends. This project helped me understand how DAX measures, date relationships, slicers and Power BI visualizations can be used together for EDA.